INCH-POUND

MIL-C-3098/60E
28 June 1995
SUPERSEDING
MIL-C-3098/60D
23 June 1975

MILITARY SPECIFICATION SHEET

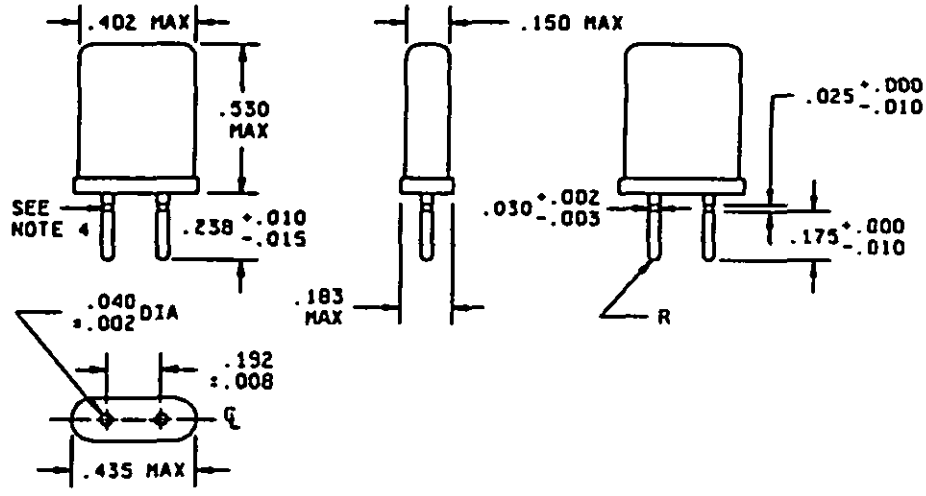
CRYSTAL UNIT, QUARTZ, CR83/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for the acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 50 to 125 MHz; fifth mechanical overtone; noncontrolled; series.

Inches	mm
.002	0.05
.003	0.08
.008	0.20
.010	0.25
.015	0.38
.025	0.64
.030	0.76
.040	1.02
.150	3.81
.175	4.45
.183	4.65
.192	4.88
.238	6.05
.402	10.21
.435	11.05
.530	13.46



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking shall be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR83/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-25/U (MIL-H-10056/11).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 50.0 to 125.0 MHz, inclusive.

Frequency tolerances:

Primary operating temperature range: ±20 ppm.

Secondary operating temperature range: ±30 ppm.

Equivalent resistance:

MHz	Ohms, maximum
50 to 90	50
90+ to 125	60

Mode of oscillation: Fifth mechanical overtone.

Operating temperature ranges:

Primary: -40°C to +90°C, inclusive.

Secondary: -55°C to -40°C, and +90°C to +105°C, inclusive.

Capacitance, shunt: 7 pF, maximum.

Shock: (specified pulse)

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Frequency and equivalent resistance (temperature run):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 204 of MIL-STD-202, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0680-10)

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